

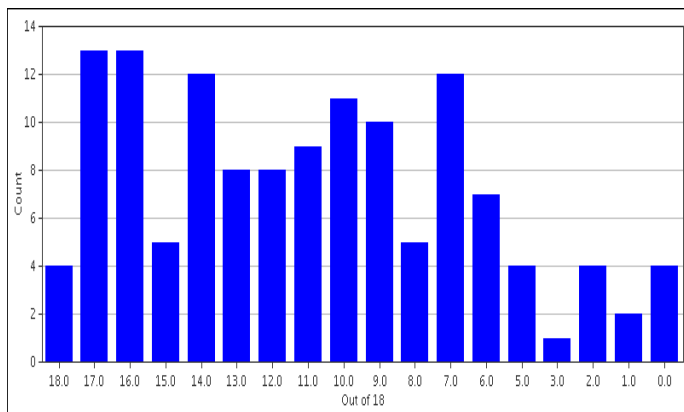
Big Data (H)

Summer 2019 Exam Feedback

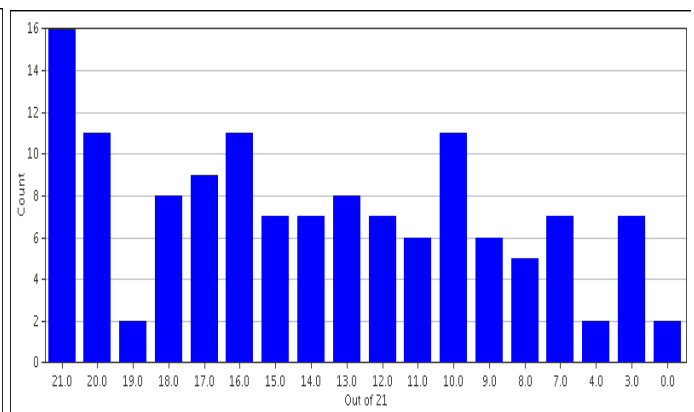
The following discusses how well you coped with the exam questions. You can find the complete answers to the exam questions on the course's Moodle page.

In general, most students did quite well, with 50/134 students (~37%) receiving a mark in the A's and another 28 students (~20%) receiving a B, and only 18/134 students (~13%) receiving a mark in the E's or below. The new format of the exam seemed to resonate with most students, with feedback so far praising it for having questions that were challenging but fair, and the multiple choice component helping students to better position their answers/explanations. This format also meant that, even if someone got the multiple choice component wrong, they could at least receive some marks through the discussion of their answer (and indeed, in some cases students received almost full marks, despite having picked the wrong MCQ answer).

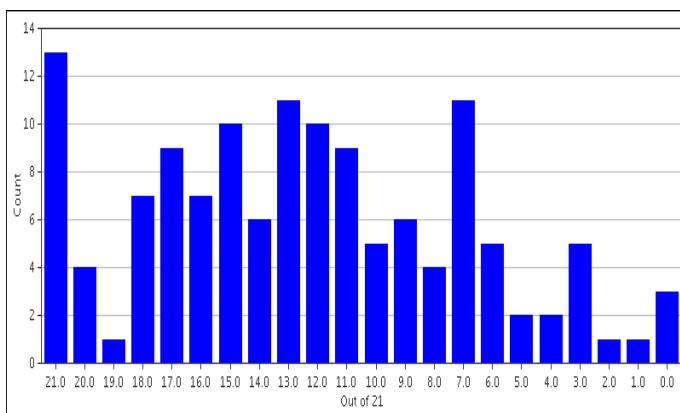
An analysis of the results showed that most students performed equally well across the 3 questions (i.e., per-question scores normalised by the question's weight where more or less at the same level across all 3 questions), with very few cases with large variations (i.e., students receiving marks differing by more than a couple of bands across questions). That said, the marks across all three questions showed a rather weak correlation (~0.36 for all three pairs); this could be an artifact of students focusing on specifics parts of the course content during their revision. Histograms and boxplots are provided for the sake of clarity and completeness for all three questions and the final exam award.



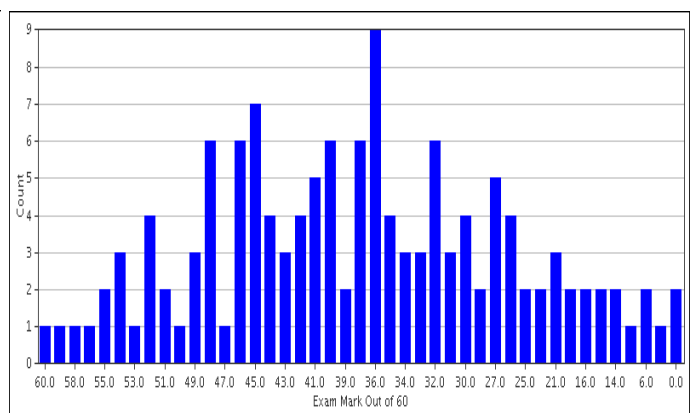
Question 1



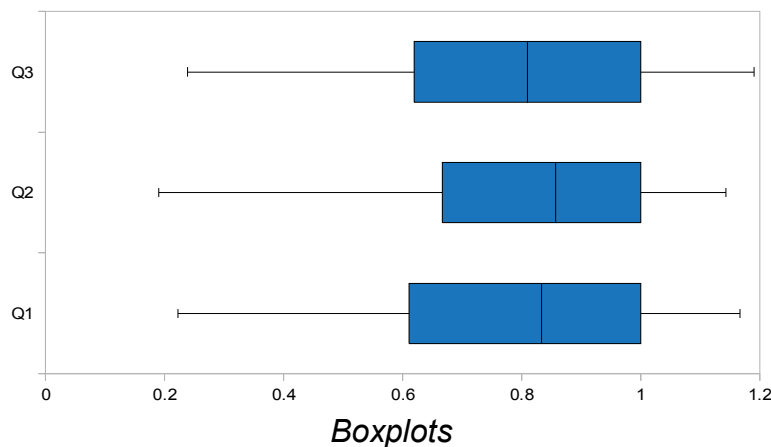
Question 2



Question 3



Cumulative



Question 1:

- A. This was a bookwork/synthesis question, asking students to pick the reason why distributed file systems like HDFS use a quite large block size, and explain their choice. Options (b) and (d) are obviously wrong, as the total data size does not change with the size of the individual block. Option (c) is also wrong; there is no write parallelism in HDFS (GFS supports concurrent appends but at different offsets) and it is safe to assume that our system handles files that are much larger than the block size and hence consist of a significant number of blocks anyway. That said, partial marks were awarded for answers that picked (b) or (d) and made the case that the per-block metadata would pose considerable overhead with a much smaller block size.
- B. This was another bookwork/synthesis question, asking students to discuss the default replica placement scheme for GFS/HDFS. Most students did quite well in this question. The parts that many students missed were (i) the fact that replication is a given (it's not a question of whether to create replicas but of where to place them – hence the disk/memory storage load on the NN does not change with the placement policy, rendering options (a) and (c) invalid), and (ii) the fact that the perhaps simplest way of assigning replicas to nodes (i.e., uniformly at random) would also lead to a balanced storage load distribution across DN's (rendering option (d) invalid). Partial marks were awarded for discussions that were explaining the advantages/design rationale of the replication scheme.

Question 2:

- A. This question was largely a problem solving/synthesis one, where students had to discuss the optimal number of reducers for a MapReduce job having full unhindered access to all cluster resources. Most students did well. That said, there were many students missed the point of the single occupancy of the cluster. Specifically, there were several students who misunderstood the statement “only one job executes at your cluster” to mean “all jobs have a parallelism factor of 1”, when it actually meant that only one job runs at a time with full access to all resources (hence option (a) would be suboptimal). There were also several students who missed the point that the number of keys in the final output cannot be known beforehand (unless a preliminary job is executed to count the number of unique words, but this would also be suboptimal given we want to execute a wordcount-type job anyway) and even if it were, it would be too high a number as explained in the question (hence option (b) is not possible/practical). Partial marks were awarded to students choosing option (c) on the basis of the network being the bottleneck.
- B. This was also a problem solving/synthesis question. Most students also did well in this question. Out of the four options, three were irrelevant as the type of join algorithm has no impact on the number of tables being joined (option (a)), the type of key used (option (b)) or the correctness of the result set (option (d)). However, several students opted for the final option (d), with rather artificial explanations; no marks were awarded in this case.
- C. This was a bookwork/problem solving question, building on both the lecture material and the students' experience acquired through the 2nd assessed exercise. Several students seemed to be confused and believing that the input RDD computation would kick in in any of the three first lines. However, Spark's lazy evaluation approach translates to RDDs being computed only when an action is found in the DAG of operations, and `count()` is the only action in all four lines of the question.
- D. This was also a bookwork/problem solving question, building on both the lecture material and the students' experience acquired through the 2nd assessed exercise. The output needs to contain a list of node IDs, hence the two options return a count of items (options (b) and (d)) are clearly wrong. The requested operation is akin to a reverse web graph, hence the edges need to be flipped and grouped

by destination node ID; hence option (c) is the correct one (option (a) would output the outlinks of every node, not its inlinks).

Question 3:

- A. This question was a mix of bookwork and problem solving. Most students did quite well in this question. The central point was that the banking application handles a fairly small amount of very important data where consistency is of utmost importance, rendering options (a) and (c) invalid. The second point is that the high-volume chat application most probably needs to be scaled out, hence partition tolerance becomes important (rendering option (b) invalid). Partial marks were awarded for answers choosing incorrect options but explaining with reasonable arguments the constraints and design considerations.
- B. This is a standard bookwork question. The vast majority of the students did quite well in this question, with those that picked an incorrect option probably not having understood the partitioning scheme well enough.
- C. This is also a standard bookwork question. In such systems, the table name and column family name are recorded as parts of the pathname of the directory under which the actual data files are stored (hence options (b), (c) and (d) are invalid). Most students that did well in part (B) of Question 3, also did well in this part.
- D. The final question of the exam paper was a bookwork/synthesis one, requiring that students know the different consistency levels and can then reason on their performance characteristics. A point that some students missed was that replication is again a given; picking option (a) or (b) won't affect the number of replicas of a data item, just the number of replicas updated at every operation. The next main point of the question is the focus on write latency (no focus on read latency). Between options (c) and (d), the latter has obviously the highest write latency as it needs to contact (and wait for acknowledgement from) all 5 replica holders, as opposed to 3 replica holders required for a quorum write. Partial marks were awarded to answers picking option (d) on the basis of its providing the required semantics.

Last, the Intended Learning Outcomes of the course were assessed by the following assessed exercises and exam questions:

- 1. *Design, employ and evaluate programs to access big data repositories in a massively parallel manner:* Assessed exercises 1 and 2, Assessed YACRS quizzes, Exam questions 2A, 2B, 2C and 2D.
- 2. *Describe and contrast the internals of the design and implementation of current cloud data storage and processing systems:* Assessed exercise 1 and 2, Assessed YACRS quizzes, All exam questions.
- 3. *Identify and discuss issues related to the scalability and efficiency challenges when processing complex queries/algorithms against big data systems, and propose ways of addressing said challenges:* Assessed exercises 1 and 2, Assessed YACRS quizzes, Exam questions 1, 2A, 2B, and 3D.
- 4. *Demonstrate that they have mastered the required background knowledge to pursue graduate studies in the fields of cloud systems and big data:* Assessed exercises 1 and 2, Assessed YACRS quizzes, All exam questions.